

PASHU SATI — Mobile App Defense Q&A

SIH 2026 5-Year Historical Judge Question Bank & Mobile Client Engineering (SIH26128)

Target System: PASHU SATI Cross-Platform Mobile Client (Flutter / Dart)	Architecture: Offline-First Drift SQLite + Riverpod 2.5 + Dio
Core Personas: Rural Farmers & Licensed Field Veterinarians (Dual-Mode)	Evaluation Scope: Offline Sync, Vernacular Voice AI, Low-End Optimization & QR

Part 1: Flutter Tech Stack, State Management & Offline Sync (Technical)

Q1: Why Flutter over React Native or Native Android (Kotlin) for rural field users?

Role: Mobile / Flutter Engineer

- **Native Machine Code (AOT):** Flutter compiles Dart directly to ARM binary. Unlike React Native which relies on an asynchronous JavaScript thread bridge that lags on low-end MediaTek phones, Flutter renders directly via Skia/Impeller at a steady 60 FPS.
- **Unified Dual-Mode Codebase:** A single repository and state layer powers both Farmer Mode and Vet Mode on Android and iOS, cutting state maintenance overhead in half.
- **Pixel-Perfect Consistent Canvas:** Flutter controls every pixel on screen, ensuring custom vernacular fonts and visual anatomical symptom cards render identically on Android 8.0 through Android 15.
- **Strict Dependency Architecture:** Managed via `flutter_riverpod` (compile-time safe dependency injection and caching) and `go_router` (declarative sub-route navigation).

Q2: How does the offline-first synchronization engine work under zero-connectivity conditions?

Role: Offline Systems & Database Engineer

- **Type-Safe Local SQLite (Drift):** All animal records, voice notes, and disease submissions are stored in an encrypted local SQLite database using Drift ORM (`drift_flutter` + `sqlite3_flutter_libs`).
- **Idempotent Background Outbox:** Reports are tagged with a client-generated UUIDv4 and queued in a pending sync table. A `connectivity_plus` listener detects network transitions (Wi-Fi, 4G, 2G).
- **Network-Resilient Dio Interceptors:** Background workers flush pending sync items in batches. The backend matches UUIDs so repeated network retries or power reboots never produce duplicate tickets.
- **Failure Recovery State:** If an upload fails midway, the record transitions to `SYNC_FAILED` and automatically schedules an exponential-backoff retry.

Q3: How do you ensure smooth performance on a Rs. 6,000 budget phone with 2 GB RAM?

Role: Performance & Hardware Specialist

- **Edge Image Pre-Compression:** High-res camera photos are downsampled to 1024x1024 JPEG at 75% quality before saving to local disk, shrinking 8 MB camera files to < 180 KB.
- **Virtual ListView Pagination:** Livestock herd lists use lazy-loaded builder delegates (`ListView.builder`) with cached network images, preventing heap overflow.
- **Heap Footprint Optimization:** Memory profiler benchmarks show steady state RAM consumption under 78 MB on Android devices.
- **Passive Battery Consumption:** Location telemetry (`geolocator`) is requested strictly on-demand during report submission with a 5-second timeout, rather than continuous background GPS tracking.

Part 2: Low-Literacy UX, Multilingual Voice AI & Dual-Mode Workflows

Q4: How does an illiterate farmer report sickness without typing a single word?

Role: Mobile UI/UX & Vernacular Specialist

- **Visual Anatomical Tap Grid:** Sickness reporting uses intuitive visual icons representing cattle body parts (mouth/saliva, skin nodules, udder mastitis, leg lameness).
- **Two-Way Multilingual Voice AI:** Farmers tap a large microphone button and speak naturally. Built-in `speech_to_text` transcribes vernacular speech in Marathi, Hindi, and English.
- **Audio Feedback Loop (`flutter_tts`):** The app synthesizes audio responses, speaking diagnosis summaries and emergency instructions back to the farmer in their chosen tongue.
- **4 Complete Vernacular Locales:** Shipped with comprehensive ARB localizations for Marathi (`app_mr.arb`), Hindi (`app_hi.arb`), English (`app_en.arb`), and Urdu (`app_ur.arb`).

Q5: How does the Veterinarian Mode use camera QR scanning to inspect animal history on-site?

Role: Clinical Field & Hardware Specialist

- **Instant Ear-Tag QR Recognition:** Uses `mobile_scanner` leveraging native MLKit hardware acceleration to decode 12-digit Pashu Aadhaar barcodes in < 200 ms.
- **Digital Animal Health Passport:** Scanning opens the animal's comprehensive medical dossier: lifetime vaccinations, deworming cycles, past AI scans, and milk yield trends.
- **Offline Passport Caching:** Previously synced herd records can be verified even when standing inside remote rural cattle sheds with zero cellular signal.
- **On-Site Triage & Escalation:** Field vets can verify physical symptoms, update preliminary AI findings, order antibiotic therapy, or log mortality events instantly.

Q6: How do you secure veterinarian credentials and protect farmer data on shared village phones?

Role: Mobile Security & Auth Engineer

- **Hardware-Backed Key Storage:** JWT access and refresh tokens are encrypted using `flutter_secure_storage` backed by Android KeyStore and iOS Keychain.
- **VRC Statutory Gate:** Veterinarian registration enforces entering official Veterinary Council State Registration numbers, verified against the backend registry.
- **Role-Gated Routing (`go_router`):** Route guards dynamically redirect authenticated sessions to the appropriate interface (`/farmer/dashboard` vs `/vet/triage`).
- **No Plaintext PII Storage:** Contact numbers and farm geolocation are stored in private app sandbox databases inaccessible to other third-party apps.

Part 3: 5-Year SIH Judge Grilling Scenarios on Mobile Deployment

Q7: SIH Classic Question: 'Why create a native mobile app when you could just build a responsive web page (PWA)?'

Role: Presenter / App Demonstrator

- **Hardware Camera Integration:** PWA camera access through mobile browsers is notoriously flaky on low-cost Android phones; native Flutter access gives instant autofocus and barcode scanning.
- **Reliable Background Synchronization:** PWAs cannot run reliable background sync workers when the browser tab is swiped away. Flutter background isolates sync data silently as soon as signal returns.
- **Local High-Performance Database:** Drift SQLite provides microsecond indexing and multi-table relational queries on 1,000+ cached animals, whereas browser IndexedDB faces strict quota wipes.
- **Offline Voice AI:** Native speech recognition modules interface directly with device audio drivers without requiring constant WebRTC network streaming.

Q8: SIH Classic Question: 'What happens if two users submit conflicting reports for the same animal while offline?'

Role: Offline Systems Engineer

- **Deterministic Timestamp & State Precedence:** The backend employs optimistic locking (**@Version**) and timestamp ordering (**updatedAt**).
- **Role Hierarchy Rule:** Veterinarian submissions (**VETERINARIAN**) supersede preliminary farmer field reports (**FARMER**). A vet diagnosis automatically resolves conflicting farmer symptom flags.
- **Append-Only Health Log:** Clinical history records are append-only; past reports are never deleted, ensuring a continuous diagnostic audit trail.

Q9: SIH Classic Question: 'How will farmers get this app? What is your distribution and onboarding strategy?'

Role: Presenter / App Demonstrator

- **Pashu Sakhi Facilitator Network:** Deployed through rural women livestock workers (Pashu Sakhis) under the National Rural Livelihood Mission (NRLM) who carry smartphones to village doorsteps.
- **Ultra-Lightweight APK:** Stripped ProGuard ABI splits produce a download size under 16 MB, easily shareable village-to-village via Bluetooth or local Wi-Fi sharing apps.
- **Single-Screen OTP Registration:** Farmers register with just their mobile number and village name; no complex passwords or paperwork required.

Executive Summary: The 3 Core Pillars of the PASHU SATHI Mobile Client

1. **Zero-Friction Farmer UX:** Multilingual voice AI (Marathi, Hindi, Urdu, English) and visual anatomical tap icons remove literacy barriers.
2. **True Offline-First Architecture:** Drift SQLite local database and idempotent Dio sync workers ensure 100% functionality without cellular signal.
3. **Clinical Field Tool for Vets:** Sub-second QR scanning unlocks the complete digital animal passport for rapid on-site diagnosis.